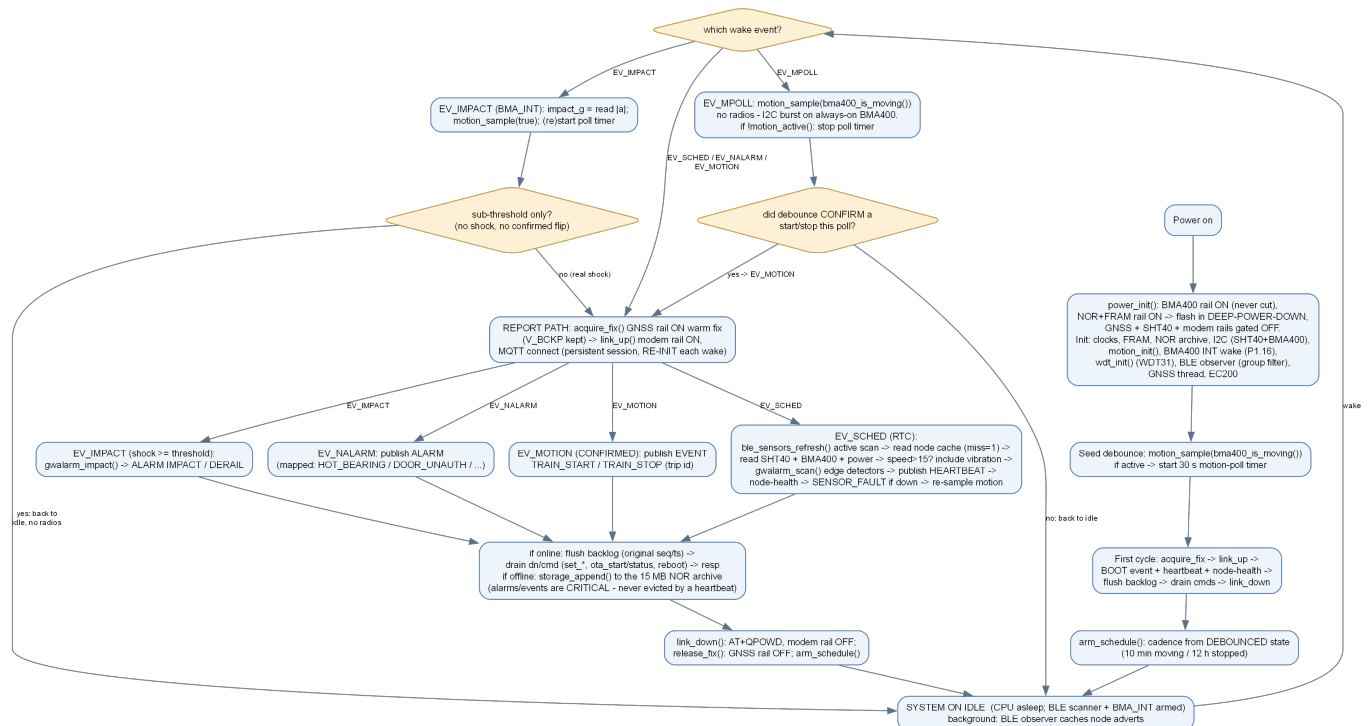


Smart Wagon Gateway — Flowchart & Algorithm

Data concentrator (Class A) · nRF54L15 + LC29HBA GNSS + EC200U 4G + BMA400/SHT40



1. Purpose & operating mode

The gateway (data concentrator) is an **uplink-first, Class-A** device: it sleeps to meet the energy budget and wakes only on the RTC schedule, a BMA400 impact interrupt, or a sensor-node alarm advert. On each wake it fixes position, gathers the latest sensor readings, publishes over the EC200U 4G modem, drains any queued commands, and returns to sleep. When the link is down, reports are written to the NOR archive and replayed later with their original `seq/t.s`.

2. Algorithm

2.1 Boot & sync

1. Init clocks (HFXO/LFXO), FRAM (config, `seq`, archive pointers), NOR archive, I2C sensors, BMA400 impact interrupt (P1.16), the WDT31 watchdog, and the low-power BLE observer with the wagon-group filter.
2. SYNC: power the GNSS rail, obtain the first fix, set the RTC from UTC, power the rail off (keep-alive stays on for warm starts).
3. **Read motion first:** from the GNSS speed decide running vs stopped, and set the initial report cadence accordingly (**10 min moving / 12 h stopped**) before going idle.
4. Connect: bring the modem up, MQTT CONNECT with a persistent session (`clean_session=0`), publish the retained status birth, subscribe `dn/cmd`.
5. Enter **System ON idle** (CPU asleep, **not** System OFF) so the BLE scanner and BMA400 interrupt stay armed and can wake the CPU.

2.2 Scheduled heartbeat (10 min running / 12 h stopped)

1. Warm GNSS fix (bounded timeout; keep last-known + `fix_valid=false` on timeout).
2. **Collect sensor-node data.** The BLE observer caches every node's adverts continuously in the background. At report time the gateway also runs a short **active scan window** to refresh any stale/missing node, then reads the cache for every **provisioned** node; a node still not heard is reported `"miss":1`.
3. Read SHT40 (temp/humidity), BMA400 motion, rail/health, backup reserve.
4. Bearing vibration health is only valid at **speed > 15 km/h** (RDSO §7.9).

5. **Node-down detection:** any provisioned node not heard within ~15 min is flagged DOWN, a `SENSOR_FAULT` alarm is sent (`st:set`), and a `clear` when it reappears. In the heartbeat that node also shows `"miss":1`.
6. Run `gwalarm_scan()` - the edge-triggered detector set in section 3.
7. Build the `hb` envelope (`seq++`, UTC epoch + derived CEP50), publish QoS1 on `up/hb`.
8. **Re-evaluate cadence** from the freshly-updated motion state before returning to idle.

2.3 Position quality (RDSO §7.17)

CEP50 and the constellation list are **derived from the receiver**, not asserted as constants. `gnss_cep_m()` computes $CEP50 \approx HDOP \times 2.5$ m from the GGA sentence and returns 99 when there is no usable fix, so the cloud can tell a good fix from a bad one. `gnss_sys_json()` builds `loc.sys` from the NMEA **talker IDs actually seen**, so it reports provenance rather than intent.

Open point: `gnss.c` documents that the LC29H does not support NavIC and never enables it, while the module's own protocol notes say it does. Resolve against the exact LC29H variant before claiming NavIC in a compliance submission.

3. Alarms & events generated by the gateway

All of these are edge-triggered in `gwalarm.c` against per-node previous state, so a condition that stays true does not re-alarm every heartbeat.

Detector	Emits	Gate / note
Impact magnitude	IMPACT / DERAIL	two-level on <code> a </code> ; DERAIL above the higher band
Bearing vibration	FLAT_WHEEL	only while speed > <code>VIBRATION_MIN_KMH</code> (§7.9)
Load	OVERLOAD, LOAD_CHANGE	alarm + state-change event
Battery	LOW_BATTERY, NODE_LOW_BATTERY	gateway cell and per-node cell
Door	DOOR_OPEN / DOOR_CLOSE	DOOR_UNAUTH is decided with geofence + motion context
Handbrake	HANDBRAKE_APPLIED / _RELEASED	HANDBRAKE_MOVING correlates state with GNSS speed
Power path	CHARGE_START / _STOP, SRC_SWITCH, BAND_CHANGE	from the BQ25798 charger
Geofence	GEOFENCE_ENTER / _EXIT	disabled: <code>GEOFENCE_RADIUS_M 0</code> , no provisioning command exists in the protocol
Link	CONN_RESTORED	on <code>gwalarm_link_up()</code>

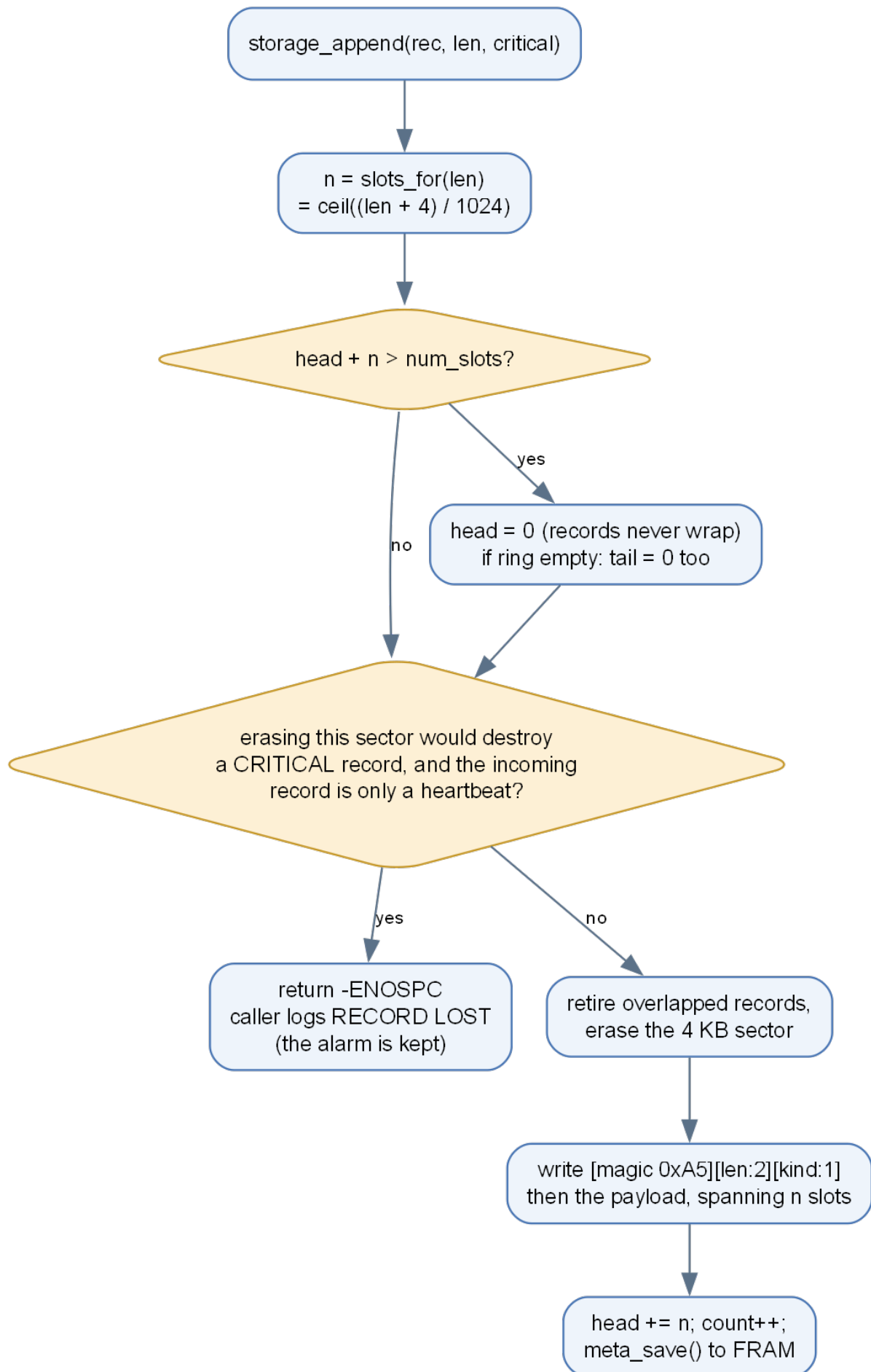
3.1 Motion debounce (a 1-min start/stop must NOT flip the state)

Transition	Must persist for	Effect
stopped → moving	<code>MOTION_START_CONFIRM_MS</code> = 2 min	confirm → <code>TRAIN_START</code> , cadence → 10 min
moving → stopped	<code>MOTION_STOP_CONFIRM_MS</code> = 5 min	confirm → <code>TRAIN_STOP</code> , cadence → 12 h

blip shorter than the above	—	pending change cancelled : no event, no cadence flip
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The always-on BMA400 is the cheap gatekeeper: while a transition is pending the gateway re-samples it every `MOTION_POLL_MS` over I2C with **no GNSS and no modem**, purely to time the persistence. Only the confirmed state is ever acted on.

4. Store-and-forward (RDSO §7.5) - "no data loss at any cost"



The archive is a ring of **1 KB slots** over the 15 MB NOR archive partition (15360 slots). **Records span as many slots as they need** - this is what lets a full ~2.4 KB heartbeat on a fully-fitted wagon be buffered at all. Head and tail are slot indices held in FRAM, so they

survive resets.

Property	Behaviour
Record header	[magic 0xA5][len:2][kind:1] - 4 bytes before the payload
Erase unit	4 KB NOR sector; slots are written individually into an already-erased sector
Critical records	An alarm/event is never evicted to make room for a heartbeat - the heartbeat is refused with -ENOSPC and the caller logs RECORD LOST
Corrupt header	resync_tail() scans forward to the next intact record, so one damaged record costs one record - not the whole backlog
Wrap	Records never wrap the end of the partition; head pads to 0, and an empty ring moves tail with it
Capacity	~106 days at the §7.4 10-minute moving cadence, against the §7.5 one-month requirement
Format change	The FRAM MAGIC is versioned; a firmware carrying the old record layout resets the ring instead of misparsing it

5. Downlink commands & OTA

After each uplink the gateway drains dn/cmd: set_threshold, set_interval, set_gnss, get_status, ota_start/ota_status, reboot - each answered with a correlated resp.

Path	Transport	Scope
Server → gateway	MQTT	thresholds, intervals, gateway firmware (MCUboot dual-slot, auto-revert)
Gateway → sub-node	BLE GATT, during the node's config window	thresholds only - one authenticated 20-byte write

Sub-node firmware update does not exist by design. There is no SMP/MCUmgr path on any node, and no phone can update one. A node's thresholds change only via its own gateway, and the write is accepted only if it decrypts under the per-wagon key, is addressed to that node, and carries a strictly higher downlink counter.

6. Watchdog

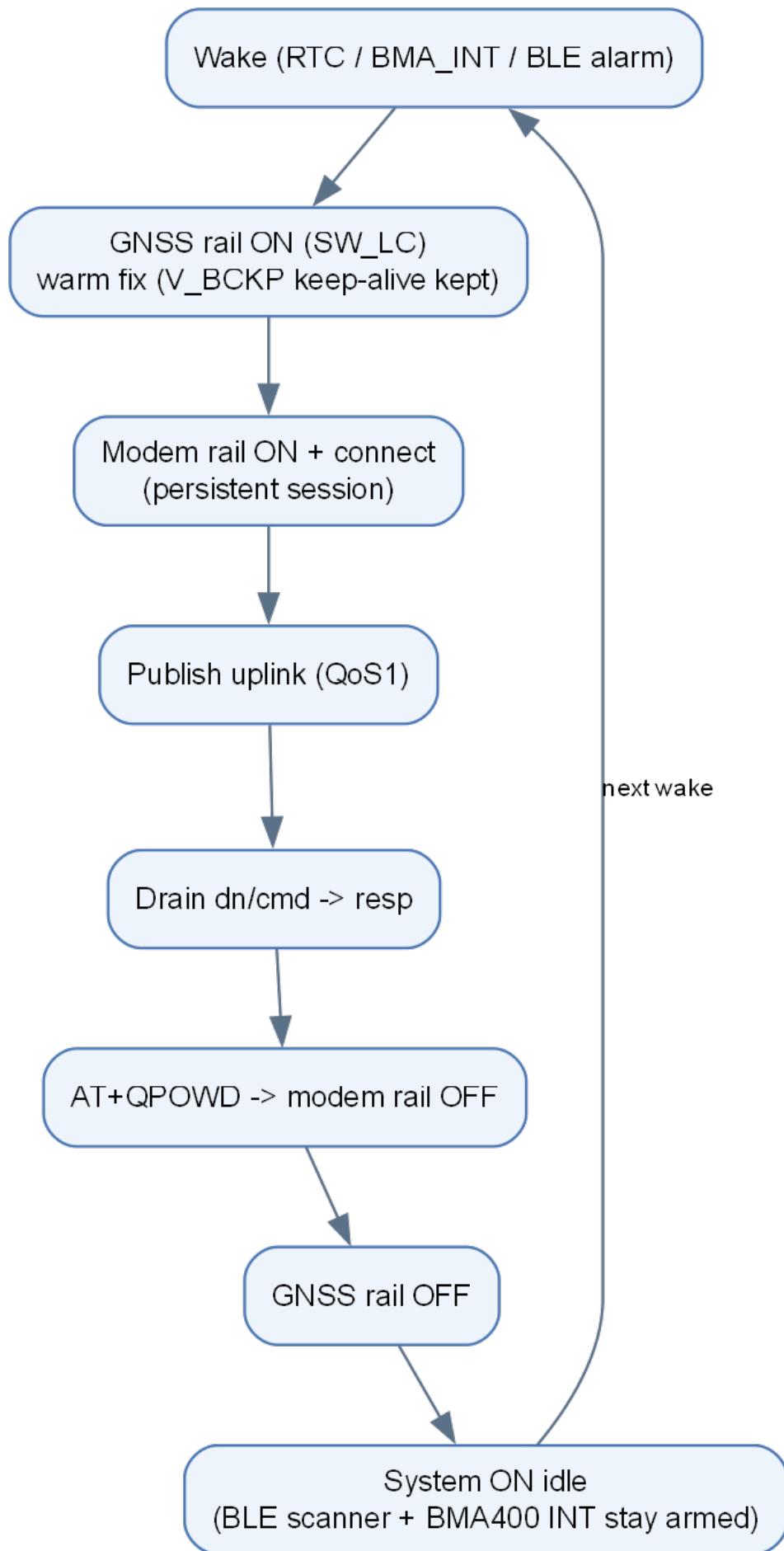
The gateway board has **no external watchdog** (the sub-nodes have a TPL5010), so the on-chip WDT31 is the only hang recovery. watchdog.c uses a **liveness gate**: the main loop must mark itself alive for the feed to happen, and long legitimate idles are bracketed with gw_wdt_idle_begin/end so sleeping is not mistaken for hanging. Timeout 120 s, feed 30 s, 10 min idle margin. Expiry causes a software reset.

7. Key parameters

Item	Value
Heartbeat cadence	10 min moving / 12 h stopped
Motion debounce	start confirm 2 min / stop confirm 5 min; poll every 30 s
Impact alarm threshold	a ≥ 4.0 g (vendor-set; §7.15 gives a 6-month calibration window)

Bearing-vibration valid speed	> 15 km/h
Critical-alarm RX window	listen 100 ms every 5 s (RX_WINDOW_MS / RX_INTERVAL_MS)
Offline buffer	15 MB NOR ring, ~106 days (§7.5 needs 1 month)
Position accuracy	CEP50 derived from HDOP; 99 reported when no fix

8. Power sequencing (per wake)



Rail (switch)	State	Powered when
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BMA400 (SW_AXI, P2.00)	always ON	impact/tamper wake source; never cut
GNSS main (SW_LC P0.01 + GNSS_PWR_EN P1.09)	gated	only around a fix; V_BCKP keep-alive stays on
EC200U modem (SW_GNSS, P2.08)	gated	only to publish; AT+QPOWD then off
SHT40 (SW_SHT_LC, P0.05)	gated	only during a climate read
NOR+FRAM (SW_MEM, P0.04)	always ON	rail kept up; flash held in deep-power-down instead

8.1 What needs re-configuring on wake

Device	Rail between reports	Loses config?	On wake
BMA400	always ON	No	Nothing - range/ODR/INT map retained
W25Q128 NOR + FRAM	always ON (DPD)	No	Release deep-power-down only
LC29H GNSS	gated (V_BCKP kept)	No (warm)	Release RESET_N; ephemeris preserved
SHT40	gated	No (stateless)	Nothing - each read is a fresh single-shot
EC200U modem	gated (off)	Yes (full)	Full re-init: CFUN/CPIN → CGATT → QIACT → QMTOPEN/QMTCONN

9. Hardware verification status

Nothing in this project has run on hardware yet. The pin map below was verified against the gateway schematic netlist; the items marked open are firmware gaps or bench checks that must be closed before production flashing.

Item	Status
All assigned nRF54L15 pins vs. schematic netlist	Verified - UARTs, I2C, SPI memory bus, 4 power switches, charger, GNSS control, BMA_INT
I2C addresses: BMA400 0x14, SHT40 0x44, BQ25798 0x6B	Verified against the schematic
GNSS_3V3 (P2.09) → TXS0102 pin 7 (VCCB)	OPEN - firmware never drives it. This GPIO is the supply for the level shifter on the EC200U UART; while it is low, that UART cannot work in either direction. Prime suspect for the silent modem.
SW_PWR (P2.10)	OPEN - no firmware use; destination not resolvable from the drawing
WP / FRAM_WP (P2.06 / P2.07)	Not driven - safe, both have 10 kΩ pull-ups so writes stay enabled
PWRKEY / RESET polarity through T1/T2	OPEN - GPIO high pulls the modem line low; verify on the bench